

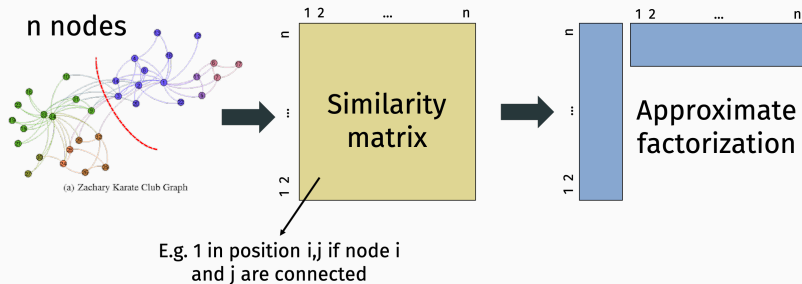
CS-GY 9223 I: Lecture 11

Spectral graph theory + randomized numerical linear algebra.

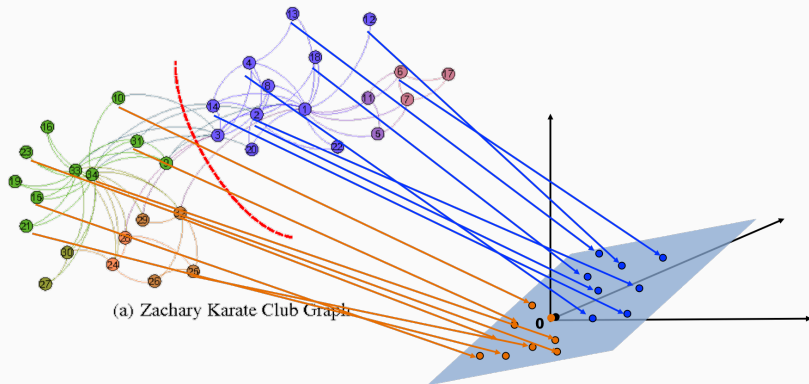
NYU Tandon School of Engineering, Prof. Christopher Musco

ENCODING GRAPH SIMILARITY

Often data is represented as a graph and similarities can be obtained from that graph:



ENCODING GRAPH SIMILARITY

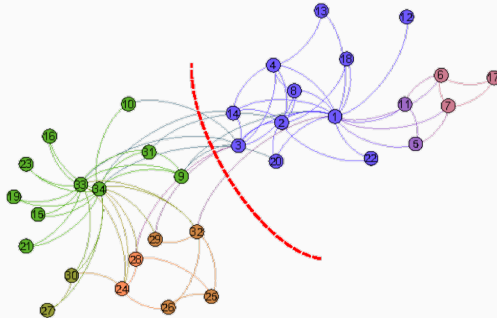


Spectral graph theory lets us formalize this heuristic idea.

CUT MINIMIZATION

Goal: Partition nodes along a cut that:

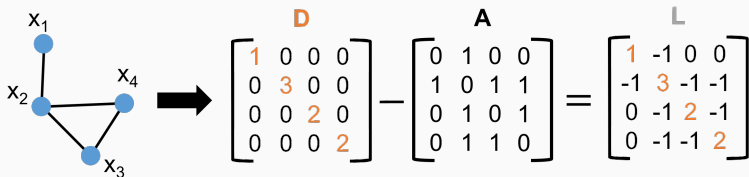
- Has few crossing edges: $|\{(u, v) \in E : u \in S, v \in T\}|$ is small.
- Separates large partitions: $|S|, |T|$ are not too small.



(a) Zachary Karate Club Graph

THE LAPLACIAN VIEW

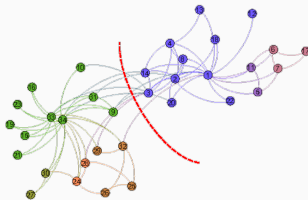
For a graph with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D}$,
 $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is the **graph Laplacian**.



$\mathbf{L} = \mathbf{B}^T \mathbf{B}$ where $\mathbf{B}$ is the “edge-vertex incidence” matrix.

$$\mathbf{B} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

THE LAPLACIAN VIEW



(a) Zachary Karate Club Graph

For a cut indicator vector $\mathbf{c} \in \{-1, 1\}^n$ with $\mathbf{c}(i) = -1$ for $i \in S$ and $\mathbf{c}(i) = 1$ for $i \in T$:

- $\mathbf{c}^T \mathbf{L} \mathbf{c} = 4 \cdot \text{cut}(S, T)$.
- $\mathbf{c}^T \mathbf{1} = |T| - |S|$.

Want to minimize both $\mathbf{c}^T \mathbf{L} \mathbf{c}$ (cut size) and $\mathbf{c}^T \mathbf{1}$ (imbalance).

Courant–Fischer min-max principle

Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ be the eigenvectors of L .

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_2 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_3 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \mathbf{v}_2} \mathbf{v}^T L \mathbf{v}$$

$$\vdots$$

$$\mathbf{v}_n = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \dots, \mathbf{v}_{n-1}} \mathbf{v}^T L \mathbf{v}$$

Courant–Fischer min-max principle

Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ be the eigenvectors of L .

$$\mathbf{v}_n = \arg \min_{\|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_{n-1} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_{n-2} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \mathbf{v}_{n-1}} \mathbf{v}^T L \mathbf{v}$$

$$\vdots$$

$$\mathbf{v}_1 = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \dots, \mathbf{v}_2} \mathbf{v}^T L \mathbf{v}$$

SMALLEST LAPLACIAN EIGENVECTOR

The smallest eigenvector/singular vector $\mathbf{v}_n$ satisfies:

$$\mathbf{v}_n = \frac{1}{\sqrt{n}} \cdot \mathbf{1} = \arg \min_{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v}$$

with $\mathbf{v}_n^T L \mathbf{v}_n = 0$.

SECOND SMALLEST LAPLACIAN EIGENVECTOR

By Courant-Fischer, $\mathbf{v}_{n-1}$ is given by:

$$\mathbf{v}_{n-1} = \underset{\|\mathbf{v}\|=1, \mathbf{v}_n^T \mathbf{v}=0}{\operatorname{argmin}} \mathbf{v}^T L \mathbf{v}$$

If $\mathbf{v}_{n-1}$ were binary, i.e. $\in \{-1, 1\}^n$, scaled by $\frac{1}{\sqrt{n}}$, it would have:

- $\mathbf{v}_{n-1}^T L \mathbf{v}_{n-1} = \operatorname{cut}(S, T)$ as small as possible **given that**
 $\mathbf{v}_{n-1}^T \mathbf{1} = |T| - |S| = 0$.
- $\mathbf{v}_{n-1}$ would indicate the smallest perfectly balanced cut.

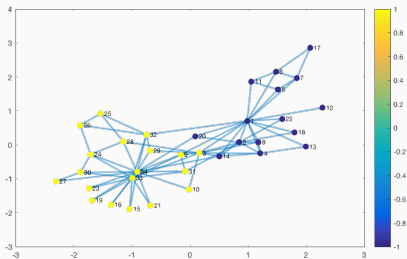
$\mathbf{v}_{n-1} \in \mathbb{R}^n$ is not generally binary, but still satisfies a ‘relaxed’ version of this property.

CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

Find a good partition of the graph by using an eigendecomposition to compute

$$\mathbf{v}_{n-1} = \arg \min_{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1}=0} \mathbf{v}^T L \mathbf{v}$$

Set S to be all nodes with $\mathbf{v}_{n-1}(i) < 0$, and T to be all with $\mathbf{v}_{n-1}(i) \geq 0$.

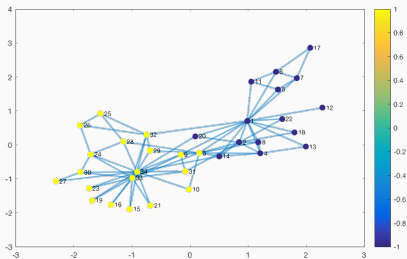


CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

Find a good partition of the graph by using an eigendecomposition to compute

$$\mathbf{v}_{n-1} = \underset{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1} = 0}{\operatorname{arg\,min}} \quad \mathbf{v}^T L \mathbf{v}$$

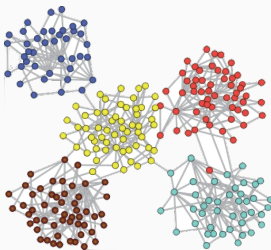
Set S to be all nodes with $\mathbf{v}_{n-1}(i) < 0$, and T to be all with $\mathbf{v}_{n-1}(i) \geq 0$.



SPECTRAL PARTITIONING IN PRACTICE

The Shi-Malik normalized cuts algorithm is one of the most commonly used variants of this approach, using the normalized Laplacian $\bar{\mathbf{L}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$.

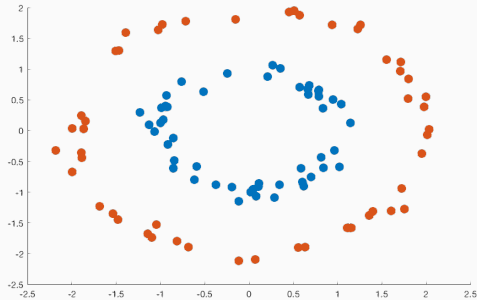
Important consideration: What to do when we want to split the graph into more than two parts?



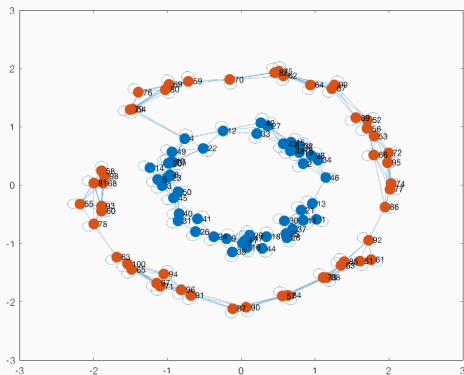
Spectral Clustering:

- Compute smallest k eigenvectors $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$ of $\mathbf{L}$.
- Represent each node by its corresponding row in $\mathbf{V} \in \mathbb{R}^{n \times k}$ whose rows are $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$.
- Cluster these rows using k -means clustering (or really any clustering method).

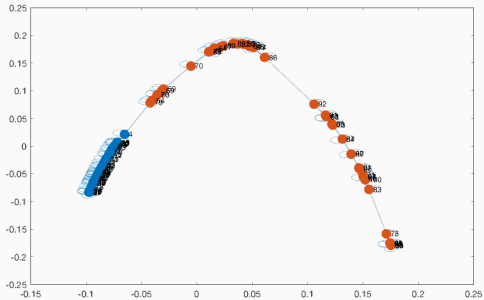
Original Data: (not linearly separable)



k -Nearest Neighbors Graph:



Embedding with eigenvectors $\mathbf{v}_{n-1}, \mathbf{v}_{n-2}$: (linearly separable)



So far: Spectral clustering partitions a graph along a small cut between large pieces.

- No formal guarantee on the ‘quality’ of the partitioning.
- Would be difficult to analyze for general input graphs.

Common approach: Give a natural **generative model** for which produces random but realistic inputs and analyze how the algorithm performs on inputs drawn from this model.

- Very common in algorithm design for data analysis/machine learning (can be used to justify ℓ_2 linear regression, k -means clustering, PCA, etc.)

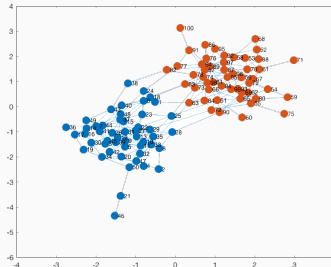
Ideas for a generative model for graphs that would allow us to understand partitioning?

STOCHASTIC BLOCK MODEL

Stochastic Block Model (Planted Partition Model):

Let $G_n(p, q)$ be a distribution over graphs on n nodes, split equally into two groups B and C , each with $n/2$ nodes.

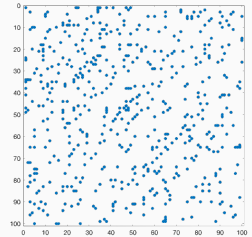
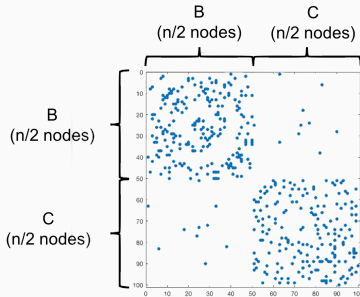
- Any two nodes in the **same group** are connected with probability p (including self-loops).
- Any two nodes in **different groups** are connected with prob. $q < p$.



LINEAR ALGEBRAIC VIEW

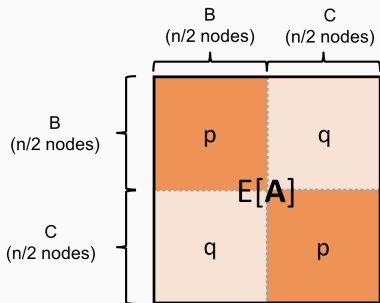
Let G be a stochastic block model graph drawn from $G_n(p, q)$.

- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be the adjacency matrix of G . What is $\mathbb{E}[\mathbf{A}]$?



EXPECTED ADJACENCY SPECTRUM

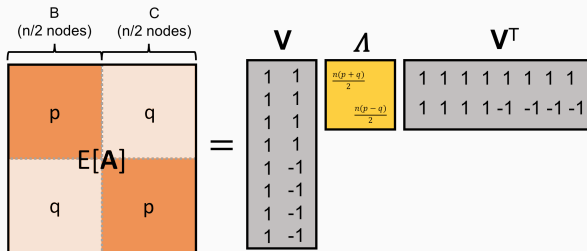
Letting G be a stochastic block model graph drawn from $G_n(p, q)$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix. $(\mathbb{E}[\mathbf{A}])_{i,j} = p$ for i, j in same group, $(\mathbb{E}[\mathbf{A}])_{i,j} = q$ otherwise.



What are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{A}]$?

Letting G be a stochastic block model graph drawn from $G_n(p, q)$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix, what are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{A}]$?

EXPECTED ADJACENCY SPECTRUM



- $\mathbf{v}_1 = \mathbf{v}_1$ with eigenvalue $\lambda_1 = \frac{(p+q)n}{2}$.
- $\mathbf{v}_2 = \chi_{B,C}$ with eigenvalue $\lambda_2 = \frac{(p-q)n}{2}$.
- $\chi_{B,C}(i) = 1$ if $i \in B$ and $\chi_{B,C}(i) = -1$ for $i \in C$.

If we compute $\mathbf{v}_2$ then we recover the communities B and C !

EXPECTED LAPLACIAN SPECTRUM

Letting G be a stochastic block model graph drawn from $G_n(p, q)$, $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix and $\mathbf{L}$ be its Laplacian, what are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{L}]$?

Upshot: The second small eigenvector of $\mathbb{E}[\mathbf{L}]$ is $\chi_{B,C}$ – the indicator vector for the cut between the communities.

- If the random graph G (equivilantly $\mathbf{A}$ and $\mathbf{L}$) were exactly equal to its expectation, partitioning using this eigenvector would exactly recover communities B and C .

How do we show that a matrix (e.g., $\mathbf{A}$) is close to its expectation? **Matrix concentration inequalities.**

- Analogous to scalar concentration inequalities like Markovs, Chebyshevs, Bernsteins.

Matrix Concentration Inequality: If $p \geq O\left(\frac{\log^4 n}{n}\right)$, then with high probability

$$\|\mathbf{A} - \mathbb{E}[\mathbf{A}]\|_2 \leq O(\sqrt{pn}).$$

where $\|\cdot\|_2$ is the matrix **spectral** norm (operator norm).

For $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\|\mathbf{X}\|_2 = \max_{\mathbf{z} \in \mathbb{R}^d: \|\mathbf{z}\|_2=1} \|\mathbf{X}\mathbf{z}\|_2$.

Exercise: Show that $\|\mathbf{X}\|_2$ is equal to the largest singular value of $\mathbf{X}$. For symmetric $\mathbf{X}$ (like $\mathbf{A} - \mathbb{E}[\mathbf{A}]$) show that it is equal to the magnitude of the largest magnitude eigenvalue.

For the stochastic block model application, we want to show that the second eigenvectors of $\mathbf{A}$ and $\mathbb{E}[\mathbf{A}]$ are close. How does this relate to their difference in spectral norm?

Davis-Kahan Eigenvector Perturbation Theorem: Suppose $\mathbf{A}, \bar{\mathbf{A}} \in \mathbb{R}^{d \times d}$ are symmetric with $\|\mathbf{A} - \bar{\mathbf{A}}\|_2 \leq \epsilon$ and eigenvectors $v_1, v_2, \dots, v_d$ and $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_d$. Letting $\theta(v_i, \bar{v}_i)$ denote the angle between v_i and $\bar{v}_i$, for all i :

$$\sin[\theta(v_i, \bar{v}_i)] \leq \frac{\epsilon}{\min_{j \neq i} |\lambda_i - \lambda_j|}$$

where $\lambda_1, \dots, \lambda_d$ are the eigenvalues of $\bar{\mathbf{A}}$.

The error gets larger if there are eigenvalues with similar magnitudes.

EIGENVECTOR PERTURBATION

$$\begin{array}{c} \mathbf{A} \\ \boxed{\begin{array}{cc} 1+\varepsilon & 0 \\ 0 & 1 \end{array}} \end{array} - \begin{array}{c} \bar{\mathbf{A}} \\ \boxed{\begin{array}{cc} 1 & 0 \\ 0 & 1+\varepsilon \end{array}} \end{array} = \begin{array}{c} \mathbf{A}-\bar{\mathbf{A}} \\ \boxed{\begin{array}{cc} \varepsilon & 0 \\ 0 & \varepsilon \end{array}} \end{array}$$

APPLICATION TO STOCHASTIC BLOCK MODEL

Claim 1 (Matrix Concentration): For $p \geq O\left(\frac{\log^4 n}{n}\right)$,

$$\|\mathbf{A} - \mathbb{E}[\mathbf{A}]\|_2 \leq O(\sqrt{pn}).$$

Claim 2 (Davis-Kahan): For $p \geq O\left(\frac{\log^4 n}{n}\right)$,

$$\sin \theta(v_2, \bar{v}_2) \leq \frac{O(\sqrt{pn})}{\min_{j \neq i} |\lambda_i - \lambda_j|} \leq \frac{O(\sqrt{pn})}{(p-q)n/2} = o\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$$

Recall: $\mathbb{E}[\mathbf{A}]$, has eigenvalues $\lambda_1 = \frac{(p+q)n}{2}$, $\lambda_2 = \frac{(p-q)n}{2}$, $\lambda_i = 0$ for $i \geq 3$.

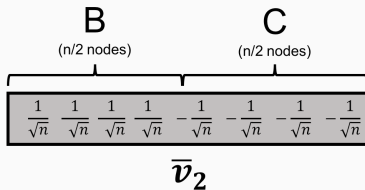
$$\min_{j \neq i} |\lambda_i - \lambda_j| = \min\left(qn, \frac{(p-q)n}{2}\right).$$

Assume $\frac{(p-q)n}{2}$ will be the minimum of these two gaps.

APPLICATION TO STOCHASTIC BLOCK MODEL

So Far: $\sin \theta(v_2, \bar{v}_2) \leq O\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$. What does this give us?

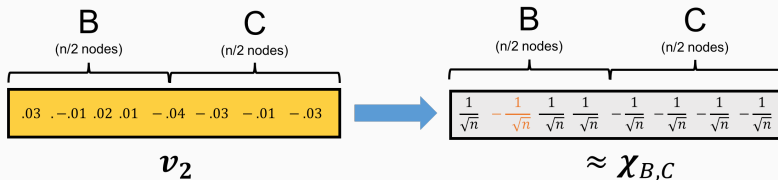
- Can show that this implies $\|v_2 - \bar{v}_2\|_2^2 \leq O\left(\frac{p}{(p-q)^2 n}\right)$ (exercise).
- $\bar{v}_2$ is $\frac{1}{\sqrt{n}}\chi_{B,C}$: the community indicator vector.



- Every i where $v_2(i)$, $\bar{v}_2(i)$ differ in sign contributes $\geq \frac{1}{n}$ to $\|v_2 - \bar{v}_2\|_2^2$.
- So they differ in sign in at most $O\left(\frac{p}{(p-q)^2}\right)$ positions.

APPLICATION TO STOCHASTIC BLOCK MODEL

Upshot: If G is a stochastic block model graph with adjacency matrix A , if we compute its second large eigenvector v_2 and assign nodes to communities according to the sign pattern of this vector, we will correctly assign all but $O\left(\frac{p}{(p-q)^2}\right)$ nodes.



- Why does the error increase as q gets close to p ?
- Even when $p - q = O(1/\sqrt{n})$, assign all but an $O(n)$ fraction of nodes correctly. E.g., assign 99% of nodes correctly.

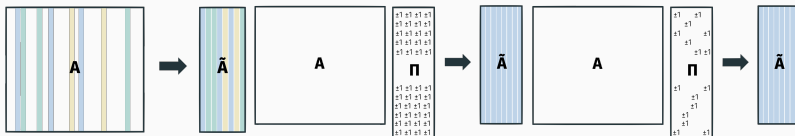
Forget about the previous problem, but still consider the matrix $\mathbf{M} = \mathbb{E}[\mathbf{A}]$.

- Dense $n \times n$ matrix.
- Computing top eigenvectors takes $\approx O(n^2/\sqrt{\epsilon})$ time.

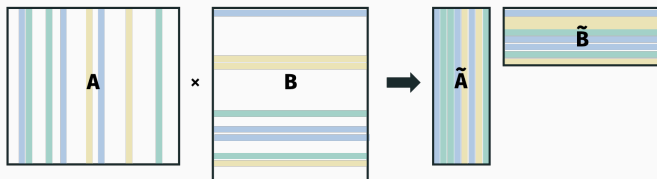
If someone asked you to speed this up and return approximate top eigenvectors, what could you do?.

Main idea: If you want to compute singular vectors or eigenvectors, multiply two matrices, solve a regression problem, etc.:

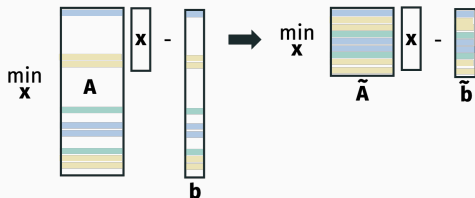
1. Compress your matrices using a randomized method.
2. Solve the problem on the smaller or sparser matrix.
 - $\tilde{A}$ called a “sketch” or “coreset” for A .



Approximate matrix multiplication:



Approximate regression:

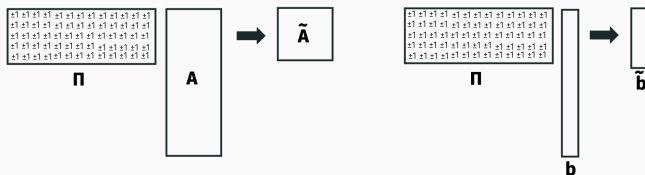


COMPARISON

	Direct	Iterative	Randomized
Method:			
Speed:			
Accuracy:			

SKETCHED REGRESSION

Randomized approximate regression using a Johnson-Lindenstrauss Matrix:



Input: $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^n$.

Goal: Want $\min_x \|\Pi A x - \Pi b\|_2^2 \leq (1 + \epsilon) \min_x \|A x - b\|_2^2$

Claim: Suffices to prove that for all $x \in \mathbb{R}^d$,

$$\|A x - b\|_2^2 \leq \|\Pi A x - \Pi b\|_2^2 \leq (1 + \epsilon) \|A x - b\|_2^2$$

Lemma (Distributional JL)

If Π is chosen to a properly scaled random Gaussian matrix, sign matrix, sparse random matrix, etc., with $O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$ rows then for any fixed $\mathbf{y}$,

$$\|\mathbf{y}\|_2^2 \leq \|\Pi\mathbf{y}\|_2^2 \leq (1 + \epsilon)\|\mathbf{y}\|_2^2$$

with probability $(1 - \delta)$.

Corollary: For any fixed $\mathbf{x}$, with probability $(1 - \delta)$,

$$\|\mathbf{Ax} - \mathbf{b}\|_2^2 \leq \|\Pi\mathbf{Ax} - \Pi\mathbf{b}\|_2^2 \leq (1 + \epsilon)\|\mathbf{Ax} - \mathbf{b}\|_2^2.$$

How do we go from “for any fixed $\mathbf{x}$ ” to “for all $\mathbf{x} \in \mathbb{R}^d$ ”.

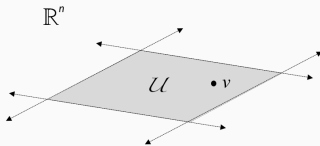
This statement requires establishing a Johnson-Lindenstrauss type bound for an infinity of possible vectors $(\mathbf{Ax} - \mathbf{b})$, which obviously can't be tackled with a union bound argument.

Theorem (Subspace Embedding from JL)

Let $\mathcal{U} \subset \mathbb{R}^n$ be a d -dimensional linear subspace in $\mathbb{R}^n$. If $\Pi \in \mathbb{R}^{m \times d}$ is chosen from any distribution $\mathcal{D}$ satisfying the Dist. JL Lemma, then with probability $1 - \delta$,

$$(1 - \epsilon)\|\mathbf{v}\|_2 \leq \|\Pi\mathbf{v}\|_2 \leq (1 + \epsilon)\|\mathbf{v}\|_2$$

for all $\mathbf{v} \in \mathcal{U}$, as long as $m = O\left(\frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right)^1$.



¹It's possible to obtain a slightly tighter bound of $O\left(\frac{d + \log(1/\delta)}{\epsilon^2}\right)$. It's a nice challenge to try proving this.

SUBSPACE EMBEDDING TO APPROXIMATE REGRESSION

Corollary: If we choose Π and properly scale then, with $O(d/\epsilon^2)$ rows, then with high probability,

$$\|\mathbf{Ax} - \mathbf{b}\|_2^2 \leq \|\Pi\mathbf{Ax} - \Pi\mathbf{b}\|_2^2 \leq (1 + \epsilon)\|\mathbf{Ax} - \mathbf{b}\|_2^2$$

for all $\mathbf{x}$ and thus

$$\min_{\mathbf{x}} \|\Pi\mathbf{Ax} - \Pi\mathbf{b}\|_2^2 \leq (1 + \epsilon) \min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2^2.$$

i.e. we can solve linear regression approximately using the $O(d/\epsilon^2) \times d$ matrix $\Pi\mathbf{A}$ in place of $\mathbf{A}$.

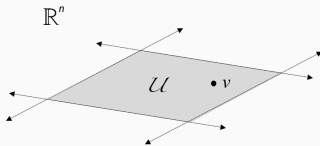
Proof: Apply Subspace Embedding Thm. to the $(d + 1)$ dimensional subspace spanned by $\mathbf{A}$'s d columns and $\mathbf{b}$. Every vector $\mathbf{Ax} - \mathbf{b}$ lies in this subspace.

Theorem (Subspace Embedding from JL)

Let $\mathcal{U} \subset \mathbb{R}^n$ be a d -dimensional linear subspace in $\mathbb{R}^n$. If $\Pi \in \mathbb{R}^{m \times d}$ is chosen from any distribution $\mathcal{D}$ satisfying the Dist. JL Lemma, then with probability $1 - \delta$,

$$(1 - \epsilon)\|\mathbf{v}\|_2 \leq \|\Pi\mathbf{v}\|_2 \leq (1 + \epsilon)\|\mathbf{v}\|_2 \quad (1)$$

for all $\mathbf{v} \in \mathcal{U}$, as long as $m = O\left(\frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right)^2$.



²It's possible to obtain a slightly tighter bound of $O\left(\frac{d + \log(1/\delta)}{\epsilon^2}\right)$. It's a nice challenge to try proving this.

Observation: The theorem holds as long as (1) holds for all $\mathbf{w}$ on the unit sphere in $\mathcal{U}$. Denote the sphere $S_{\mathcal{U}}$:

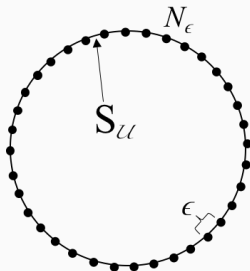
$$S_{\mathcal{U}} = \{\mathbf{w} \mid \mathbf{w} \in \mathcal{U} \text{ and } \|\mathbf{w}\|_2 = 1\}.$$

Follows from linearity: Any point $\mathbf{v} \in \mathcal{U}$ can be written as $c\mathbf{w}$ for some scalar c and some point $\mathbf{w} \in S_{\mathcal{U}}$.

- If $(1 - \epsilon)\|\mathbf{w}\|_2 \leq \|\Pi\mathbf{w}\|_2 \leq (1 + \epsilon)\|\mathbf{w}\|_2$.
- then $c(1 - \epsilon)\|\mathbf{w}\|_2 \leq c\|\Pi\mathbf{w}\|_2 \leq c(1 + \epsilon)\|\mathbf{w}\|_2$,
- and thus $(1 - \epsilon)\|c\mathbf{w}\|_2 \leq \|\Pi c\mathbf{w}\|_2 \leq (1 + \epsilon)\|c\mathbf{w}\|_2$.

SUBSPACE EMBEDDING PROOF

Intuition: There are not too many “different” points on a d -dimensional sphere:



N_ϵ is called an “ ϵ ”-net.

If we can prove

$$(1 - \epsilon) \leq \|\Pi \mathbf{w}\|_2 \leq (1 + \epsilon)$$

for all points $\mathbf{w} \in N_\epsilon$, we can hopefully extend to all of S_U .

Lemma (ϵ -net for the sphere)

For any $\epsilon \leq 1$, there exists a set $N_\epsilon \subset S_{\mathcal{U}}$ with $|N_\epsilon| = \left(\frac{4}{\epsilon}\right)^d$ such that $\forall \mathbf{v} \in S_{\mathcal{U}}$,

$$\min_{\mathbf{w} \in N_\epsilon} \|\mathbf{v} - \mathbf{w}\| \leq \epsilon.$$

1. Set $\delta = \left(\frac{\epsilon}{8}\right)^d$. By a union bound, with high probability, for all $\mathbf{w} \in N_\epsilon$,

$$(1 - \epsilon) \leq \|\Pi \mathbf{w}\|_2 \leq (1 + \epsilon).$$

as long as Π has $O\left(\frac{\log(1/\delta)}{\epsilon^2}\right) = O\left(\frac{d \log(1/\epsilon)}{\epsilon^2}\right)$ rows.

2. Consider any $\mathbf{v} \in S_{\mathcal{U}}$. You can check that, for some $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \dots \in N_{\epsilon}$, $\mathbf{v}$ can be written:

$$\mathbf{v} = \mathbf{w}_0 + c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \dots$$

for constants $c_1, c_2, \dots$ where $|c_i| \leq \epsilon^i$.

3. Applying triangle inequality, we have

$$\begin{aligned} \|\Pi \mathbf{v}\|_2 &= \|\Pi \mathbf{w}_0 + c_1 \Pi \mathbf{w}_1 + c_2 \Pi \mathbf{w}_2 + \dots\| \\ &\leq \|\Pi \mathbf{w}_0\| + \epsilon \|\Pi \mathbf{w}_1\| + \epsilon^2 \|\Pi \mathbf{w}_2\| + \dots \\ &\leq (1 + \epsilon) + \epsilon(1 + \epsilon) + \epsilon^2(1 + \epsilon) + \dots \\ &\leq 1 + O(\epsilon). \end{aligned}$$

4. Similarly,

$$\begin{aligned}
 \|\Pi v\|_2 &= \|\Pi w_0 + c_1 \Pi w_1 + c_2 \Pi w_2 + \dots\| \\
 &\geq \|\Pi w_0\| - \epsilon \|\Pi w_1\| - \epsilon^2 \|\Pi w_2\| - \dots \\
 &\geq (1 - \epsilon) - \epsilon(1 + \epsilon) - \epsilon^2(1 + \epsilon) - \dots \\
 &\geq 1 - O(\epsilon).
 \end{aligned}$$

So we have proven

$$1 - O(\epsilon) \leq \|\Pi v\|_2 \leq 1 + O(\epsilon)$$

for all v in $S_{\mathcal{U}}$.

Adjusting ϵ proves the Subspace Embedding theorem.

[t]

Theorem (Randomized Linear Regression)

Let Π be a properly scaled JL matrix (random Gaussian, sign, sparse random, etc.) with $O\left(\frac{d \log(1/\epsilon)}{\epsilon^2}\right)$ rows. Then with high probability, for any $\mathbf{A} \in \mathbb{R}^{n \times d}$ and $\mathbf{b} \in \mathbb{R}^n$,

$$\min_{\mathbf{x}} \|\Pi \mathbf{A} \mathbf{x} - \Pi \mathbf{b}\|_2^2 \leq (1 + \epsilon) \min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2^2$$

